Study Questions: Cognition

1. What is precision in the Active Inference framework? How is it defined mathematically?
2. How does precision relate to attention? Explain the concept of precision as neural gain.
3. Which cortical neurons are proposed to encode precision, and how?
4. Describe the role of acetylcholine in modulating precision. What happens when cholinergic function is impaired?
5. How does dopamine modulate precision on reward predictions and policy selection?
6. What is the relationship between noradrenaline and the brain's estimate of environmental volatility?
7. How does Lawson et al. (2014) explain autism in terms of aberrant precision?
8. Explain the ADHD phenotype in Active Inference terms. What specific precision imbalance is proposed?
9. How does anxiety relate to excessive precision on threat-related prediction errors?
10. What is the difference between tonic and phasic dopamine signaling in the context of precision?
11. How does the concept of precision relate to signal-to-noise ratio in sensory processing?
12. Compare the Active Inference account of attention with Posner's three-network model (alerting, orienting, executive). Are they compatible?
13. How might caffeine (which increases arousal/noradrenaline) affect precision dynamics?
14. What is the role of the pulvinar nucleus of the thalamus in precision weighting?
15. How does Active Inference explain the "cocktail party effect" (selectively attending to one voice in a noisy room)?
16. What is the proposed relationship between serotonin and interoceptive precision? How might SSRIs work through this mechanism?
17. How does precision relate to confidence? Is high precision the neural correlate of "certainty"?
18. Can excessive precision be harmful? Give examples from clinical conditions.
19. How does developmental maturation of neuromodulatory systems relate to changes in attention and cognition from childhood to adulthood?

20. Design an experiment that could test whether pharmacologically increasing acetylcholine (e.g., with donepezil) increases the precision of sensory prediction errors in a perceptual discrimination task.